Differentiating Power Functions and Polynomials

Purpose

In this lab we will learn how to differentiate polynomials like $x^3 + 6x^2 + 2x + 1$.

Pre-Lab

This pre-lab (and the lab itself) is computational and conceptual, and will not involve spreadsheets.

1. Recall that a derivative is just a fancy slope: it lets us compute slopes of curves in general. But we already knew how to compute the slopes of *some* functions before we introduced this concept!

   Given that the derivative is just a slope, what are the derivatives of the following functions?

   $$f(x) = c : f'(x) = ____ \qquad\qquad g(x) = mx + b : g'(x) = ____$$

2. Compute the following derivatives using the definition:

   $$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

   You must use the definition, and may not use any other tools we have not yet learned, even if you know them from high school!

   (a) $f(x) = x^2$
   (b) $f(x) = x^3$
   (c) $f(x) = \frac{1}{x}$

   (Hint for (c): You'll need to find a common denominator for the subtraction $\frac{1}{x+h} - \frac{1}{x}$ along the way. Also, remember that dividing by h is the same as multiplying by $\frac{1}{h}$.)

3. The *Constant Multiple Rule* says that

   $$\frac{d}{dx} cf(x) = cf'(x).$$

   ($\frac{d}{dx} f(x)$ means the derivative of $f(x)$)

   (a) Suppose that the derivative of some function $f(x)$ is

   $$f'(x) = \frac{2x}{\sqrt{x} + 7}$$

   What is the derivative of $17f(x)$?

(b) What is the derivative of $\frac{22}{x}$? (Hint: use your answer to 2(c) above.)

4. The *Sum Rule* says that if you want to take the derivative of two functions added together, you just need to find each of their derivatives and add them together:

$$\frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x)$$

(a) If $f(x)$ and $g(x)$ are two functions with $f'(x) = \sqrt{x}$, and $g'(x) = \frac{1}{\sqrt{x}+x}$, find the derivative of $f(x) + g(x)$.

(b) Let $f(x) = x^3 + \frac{1}{x}$. Find $f'(x)$. (Hint: use your answers to 2(b) and 2(c) above.)

5. Multiply out each of the following and simplify your answers. You should get a nice (and simple!) pattern.

(a) $(z - x)(z + x)$

(b) $(z - x)(z^2 + zx + x^2)$

(c) $(z - x)(z^3 + z^2x + zx^2 + x^3)$

(d) $(z - x)(z^4 + z^3x + z^2x^2 + zx^3 + x^4)$

Main Lab

Recasting the Derivative

6. By letting $z = x + h$ and carefully thinking about what z approaches as h approaches 0, show that

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}.$$

Differentiating Monomials - The Power Rule

A function of the form $f(x) = ax^r$ is called a *monomial*. In this part of the lab, we will learn how to find its derivative.

7. Write out each of the following a product of $z - x$ and another polynomial (use your answer to the last question from the pre-lab here.)

   (a) $z^2 - x^2 = (z - x)$_____

   (b) $z^3 - x^3 = (z - x)$_____

   (c) $z^4 - x^4 = (z - x)$_____

   (d) $z^5 - x^5 = (z - x)$_____

   (e) $z^n - x^n = (z - x)$_____

8. In the factorization of $z^3 - x^3$, how many terms are in the second factor? In general, in the factorization of $z^n - x^n$, how many terms are in the second factor?

9. Now use your factorization of $z^n - x^n$ above to find $\frac{d}{dx}(x^n) = \lim_{z \to x} \frac{z^n - x^n}{z - x}$.

Record your result here:

The Power Rule

If $f(x) = x^n$, then $f'(x) =$ _____.

10. In the pre-lab, you showed that

$$\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$$

You can similarly show that

$$\frac{d}{dx}\left(\sqrt{x}\right) = \frac{1}{2\sqrt{x}} \text{ and } \frac{d}{dx}\left(\frac{1}{x^2}\right) = -\frac{2}{x^3}$$

Rewrite each of these differentiation formulas using exponential notation (the first is done for you as an example):

(a) $\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2} \Leftrightarrow \frac{d}{dx}(x^{-1}) = -x^{-2}$

(b) $\frac{d}{dx}\left(\sqrt{x}\right) = \frac{1}{2\sqrt{x}} \Leftrightarrow \frac{d}{dx}_____ = _____$

(c) $\frac{d}{dx}\left(\frac{1}{x^2}\right) = -\frac{2}{x^3} \Leftrightarrow \frac{d}{dx}_____ = _____$

11. In Question 9 above, we showed that we have $\frac{d}{dx}(x^n) = nx^{n-1}$.

But there's a caveat here! The factorization we used to get this result (in Question 7) only works if n is a positive whole number.

Nonetheless, use the power rule to compute the derivatives of:

(a) $f(x) = \frac{1}{x}$ (aka x^{-1}).

(b) $f(x) = \frac{1}{x^2}$ (aka x^{-2}).

(c) $f(x) = \sqrt{x}$ (aka $x^{\frac{1}{2}}$).

If you do this right, you should get the same results as in the previous question. The power rule works for any exponent! We'll prove that later in the semester.

Differentiating Polynomials and Applications

12. Use these rules (and only these rules, no others!) to find the derivatives of the following functions:

(a) $y = -3x^4 - 4x^3 - 6x + 2$

(b) $y = 3t^5 - 5\sqrt{t} + \frac{7}{t}$

(c) $y = z^2 + \frac{1}{2z}$

(d) $y = \frac{x^2+1}{x}$

(e) $y = (x+1)^2$

13. Find the equation of the tangent line to the graph of $y = 2x^3 - 2x^2 + 1$ at $(1, 1)$.

14. Suppose that $f(x) = x^{10}$. What is the smallest number n such that $f^{(n)}(x) = 0$? (Recall that $f^{(n)}$ means the nth derivative.)

15. Find values of a and b such that the line $3x + y = a$ is tangent to the graph of $f(x) = bx^2$ at $x = 1$.

16. If n is even then the derivative of x^n is (choose one):

    (a) an even function (b) an odd function (c) neither even nor odd (d) not enough information

    Draw a graph of a typical graph of a function of the form $f(x) = x^n$ for even n, and add in tangent lines to your graph at $x = a$ and $x = -a$. Use your graph and tangent lines to explain your answer above.

17. If n is odd then the derivative of x^n is (choose one):

    (a) an even function (b) an odd function (c) neither even nor odd (d) not enough information

    Draw a graph of a typical graph of a function of the form $f(x) = x^n$ for odd n, and add in tangent lines to your graph at $x = a$ and $x = -a$. Use your graph and tangent lines to explain your answer above.

18. Consider the function $y = x^5 + 2x$. What is the smallest slope that a line tangent to this curve can have?

Report

For the report, each group should hand in answers to the following questions: 7, 9, 13, 15, 17.